

Project Title: E-Commerce Customer Churn Analysis

Data Cleaning: (5 marks)

Handling Missing Values and Outliers:

➤ Impute mean for the following columns, and round off to the nearest integer if required: WarehouseToHome, HourSpendOnApp, OrderAmountHikeFromlastYear, DaySinceLastOrder.

```
5664 • UPDATE customer_churn
5665   SET WarehouseToHome = (
5666     SELECT ROUND(AVG(WarehouseToHome))
5667     FROM (
5668       SELECT WarehouseToHome
5669       FROM customer_churn
5670     ) AS temp
5671   )
5672   WHERE WarehouseToHome IS NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

count(*)
0

Result Grid

```
5684 • UPDATE customer_churn
5685   SET OrderAmountHikeFromlastYear = (
5686     SELECT ROUND(AVG(OrderAmountHikeFromlastYear))
5687     FROM (
5688       SELECT OrderAmountHikeFromlastYear
5689       FROM customer_churn
5690     ) AS temp
5691   )
5692   WHERE OrderAmountHikeFromlastYear IS NULL;
5693
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

count(*)
0

```

5674 • UPDATE customer_churn
5675 • SET HourSpendOnApp = (
5676 •     SELECT ROUND(AVG(HourSpendOnApp))
5677 •     FROM (
5678 •         SELECT HourSpendOnApp
5679 •         FROM customer_churn
5680 •     ) AS temp
5681 • )
5682 • WHERE HourSpendOnApp IS NULL;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

count(*)
0

Result Grid

```

5694 • select count(*) from customer_churn where WarehouseToHome is NULL;
5695 • select count(*) from customer_churn where HourSpendOnApp is NULL;
5696 • select count(*) from customer_churn where OrderAmountHikeFromLastYear is NULL;
5697

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

count(*)
0

Dealing with Inconsistencies:

➤ Replace occurrences of “Phone” in the 'PreferredLoginDevice' column and “Mobile” in the 'PreferredOrderCat' column with “Mobile Phone” to ensure uniformity.

```

5703 • UPDATE customer_churn
5704 • SET PreferredOrderCat = 'Mobile Phone'
5705 • WHERE PreferredOrderCat = 'Mobile';

```

Output

#	Time	Action	Message	Duration / Fetch
18	13:35:00	UPDATE customer_churn SET PreferredLoginDevice = 'Mobile Phone' WHERE PreferredLoginDevice = 'Phone'	1231 row(s) affected Rows matched: 1231 Changed: 1231 Warnings: 0	0.031 sec
19	13:35:48	UPDATE customer_churn SET PreferredOrderCat = 'Mobile Phone' WHERE PreferredOrderCat = 'Mobile'	809 row(s) affected Rows matched: 809 Changed: 809 Warnings: 0	0.032 sec

5699	•	UPDATE customer_churn			
5700		SET PreferredLoginDevice = 'Mobile Phone'			
5701		WHERE PreferredLoginDevice = 'Phone';			
5702					

#	Time	Action	Message	Duration / Fetch
17	13:33:17	select * from customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
18	13:35:00	UPDATE customer_churn SET PreferredLoginDevice = 'Mobile Phone' WHERE PreferredLoginDevice = 'Phone';	1231 row(s) affected Rows matched: 1231 Changed: 1231 Warnings: 0	0.031 sec

➤ Standardize payment mode values: Replace "COD" with "Cash on Delivery" and "CC" with "Credit Card" in the PreferredPaymentMode column.

5707	•	UPDATE customer_churn			
5708		SET PreferredPaymentMode = 'Cash on Delivery'			
5709		WHERE PreferredPaymentMode = 'COD';			
5710					

#	Time	Action	Message	Duration / Fetch
19	13:35:48	UPDATE customer_churn SET PreferredOrderCat = 'Mobile Phone' WHERE PreferredOrderCat = 'Phone';	809 row(s) affected Rows matched: 809 Changed: 809 Warnings: 0	0.032 sec
20	13:36:18	UPDATE customer_churn SET PreferredPaymentMode = 'Cash on Delivery' WHERE PreferredPaymentMode = 'COD';	365 row(s) affected Rows matched: 365 Changed: 365 Warnings: 0	0.016 sec

5711	•	UPDATE customer_churn			
5712		SET PreferredPaymentMode = 'Credit Card'			
5713		WHERE PreferredPaymentMode = 'CC';			

#	Time	Action	Message	Duration / Fetch
20	13:36:18	UPDATE customer_churn SET PreferredPaymentMode = 'Cash on Delivery' WHERE PreferredPaymentMode = 'COD';	365 row(s) affected Rows matched: 365 Changed: 365 Warnings: 0	0.016 sec
21	13:36:44	UPDATE customer_churn SET PreferredPaymentMode = 'Credit Card' WHERE PreferredPaymentMode = 'CC';	273 row(s) affected Rows matched: 273 Changed: 273 Warnings: 0	0.047 sec

Data Transformation: (3 marks)

Column Renaming :

➤ Rename the column "PreferedOrderCat" to "PreferredOrderCat".

5724	•	ALTER TABLE customer_churn			
5725		RENAME COLUMN PreferredOrderCat TO PreferredOrderCat;			
5726					
5727					

#	Time	Action	Message	Duration / Fetch
24	13:38:25	SELECT DISTINCT PreferredPaymentMode FROM customer_churn LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
25	13:42:08	ALTER TABLE customer_churn RENAME COLUMN PreferredOrderCat TO PreferredOrderCat;	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.015 sec

➤ Rename the column "HourSpendOnApp" to "HoursSpentOnApp".

```
5727 • ALTER TABLE customer_churn
5728 RENAME COLUMN HourSpendOnApp TO HoursSpentOnApp;
```

Output

#	Time	Action	Message	Duration / Fetch
25	13:42:08	ALTER TABLE customer_churn RENAME COLUMN PreferredOrderCat TO PreferredOr...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.015 sec
26	13:42:36	ALTER TABLE customer_churn RENAME COLUMN HourSpendOnApp TO HoursSp...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.015 sec

Creating New Columns:

➤ Create a new column named 'ComplaintReceived' with values "Yes" if the corresponding value in the 'Complain' is 1, and "No" otherwise.

```
5733 • UPDATE customer_churn
5734 SET ComplaintReceived =
5735 CASE
5736 WHEN Complain = 1 THEN 'Yes'
5737 ELSE 'No'
5738 END;
5739
5740
```

Output

#	Time	Action	Message	Duration / Fetch
28	13:44:18	ALTER TABLE customer_churn ADD ComplaintReceived VARCHAR(5)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.015 sec
29	13:44:28	UPDATE customer_churn SET ComplaintReceived = CASE WHEN Complain = 1 T...	5630 row(s) affected Rows matched: 5630 Changed: 5630 Warnings: 0	0.796 sec

➤ Create a new column named 'ChurnStatus'. Set its value to "Churned" if the corresponding value in the 'Churn' column is 1, else assign "Active".

```
5743 • ALTER TABLE customer_churn
5744 ADD ChurnStatus VARCHAR(10);
5745
5746 • UPDATE customer_churn
5747 SET ChurnStatus =
5748 CASE
5749 WHEN Churn = 1 THEN 'Churned'
5750 ELSE 'Active'
5751 END;
5752
5753 • select ChurnStatus from customer_churn
```

Result Grid

ChurnStatus
Churned
Churned
Churned
Churned
Churned

customer_churn 10 x

Column Dropping:

- Drop the columns "Churn" and "Complain" from the table.

```
5755 ALTER TABLE customer_churn
5756 DROP COLUMN Churn,
5757 DROP COLUMN Complain;
```

Output

Action Output

#	Time	Action	Message
38	13:49:52	select ChurnStatus from customer_churn LIMIT 0, 1000	1000 row(s) returned
39	13:50:11	ALTER TABLE customer_churn DROP COLUMN Churn, DROP COLUMN Complain	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

Data Exploration and Analysis: (17 marks)

- Retrieve the count of churned and active customers from the dataset.

```
5759 SELECT ChurnStatus, COUNT(*) AS CustomerCount
5760 FROM customer_churn
5761 GROUP BY ChurnStatus;
```

Result Grid

ChurnStatus	CustomerCount
Churned	948
Active	4682

- Display the average tenure and total cashback amount of customers who churned.

```
5763 SELECT
5764 AVG(Tenure) AS AverageTenure,
5765 SUM(CashbackAmount) AS TotalCashback
5766 FROM customer_churn
5767 WHERE ChurnStatus='Churned';
```

Result Grid

AverageTenure	TotalCashback
3.3795	152030

➤ Determine the percentage of churned customers who complained.

```
5769 • SELECT
5770     ROUND(
5771     COUNT(CASE WHEN ComplaintReceived='Yes' THEN 1 END)*100.0/COUNT(*),
5772     2
5773     ) AS ComplaintPercentage
5774 FROM customer_churn
5775 WHERE ChurnStatus='Churned';
```

Result Grid		Filter Rows:	Exports:	Wrap Cell Content:
ComplaintPercentage				
▶	53.59			

➤ Identify the city tier with the highest number of churned customers whose preferred order category is Laptop & Accessory.

```
5778 • SELECT CityTier, COUNT(*) AS TotalCustomers
5779 FROM customer_churn
5780 WHERE ChurnStatus='Churned'
5781 AND PreferredOrderCat='Laptop & Accessory'
5782 GROUP BY CityTier
5783 ORDER BY TotalCustomers DESC
5784 LIMIT 1;
```

➤ Identify the most preferred payment mode among active customers.

```
5786 • SELECT PreferredPaymentMode, COUNT(*) AS Total
5787 FROM customer_churn
5788 WHERE ChurnStatus='Active'
5789 GROUP BY PreferredPaymentMode
5790 ORDER BY Total DESC
5791 LIMIT 1;
```

Result Grid		Filter Rows:	Exports:	Wrap Cell Content:	Fetch rows:
PreferredPaymentMode		Total			
▶	Debit Card	1958			

➤ Calculate the total order amount hike from last year for customers who are single and prefer mobile phones for ordering.

```
5793 • SELECT SUM(OrderAmountHikeFromLastYear) AS TotalHike
5794 FROM customer_churn
5795 WHERE MaritalStatus='Single'
5796 AND PreferredLoginDevice='Mobile Phone';
```

TotalHike
19593

➤ Find the average number of devices registered among customers who used UPI as their preferred payment mode.

```
5798 • SELECT AVG(NumberOfDeviceRegistered) AS AvgDevices
5799 FROM customer_churn
5800 WHERE PreferredPaymentMode='UPI';
```

AvgDevices
3.7174

➤ Determine the city tier with the highest number of customers.

```
5802 • SELECT CityTier, COUNT(*) AS TotalCustomers
5803 FROM customer_churn
5804 GROUP BY CityTier
5805 ORDER BY TotalCustomers DESC
5806 LIMIT 1;
```

CityTier	TotalCustomers
1	3666

➤ Identify the gender that utilized the highest number of coupons.

```

5808 • SELECT Gender, SUM(CouponUsed) AS TotalCoupons
5809 FROM customer_churn
5810 GROUP BY Gender
5811 ORDER BY TotalCoupons DESC
5812 LIMIT 1;

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:	Result Grid
	Gender	TotalCoupons				
▶	Male	5497				

➤ List the number of customers and the maximum hours spent on the app in each preferred order category.

```

5815 • SELECT
5816 PreferredOrderCat,
5817 COUNT(*) AS CustomerCount,
5818 MAX(HoursSpentOnApp) AS MaxHours
5819 FROM customer_churn
5820 GROUP BY PreferredOrderCat;

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:	
	PreferredOrderCat	CustomerCount	MaxHours		
▶	Laptop & Accessory	2050	5		
	Mobile Phone	2080	5		
	Others	264	4		
	Fashion	826	5		
	Grocery	410	4		

➤ Calculate the total order count for customers who prefer using credit cards and have the maximum satisfaction score.

```

5831 • SELECT AVG(SatisfactionScore) AS AvgSatisfaction
5832 FROM customer_churn
5833 WHERE ComplaintReceived='Yes';

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:	
	AvgSatisfaction				
▶	2.9988				

➤ What is the average satisfaction score of customers who have complained?

```
5831 • SELECT AVG(SatisfactionScore) AS AvgSatisfaction
5832 FROM customer_churn
5833 WHERE ComplaintReceived='Yes';
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
AvgSatisfaction	2.9988			

➤ List the preferred order category among customers who used more than 5 coupons.

```
5836 • SELECT DISTINCT PreferredOrderCat
5837 FROM customer_churn
5838 WHERE CouponUsed>5;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
PreferredOrderCat	Others			
	Laptop & Accessory			
	Fashion			
	Mobile Phone			
	Grocery			

➤ List the top 3 preferred order categories with the highest average cashback amount.

```
5841 • SELECT
5842 PreferredOrderCat,
5843 AVG(CashbackAmount) AS AvgCashback
5844 FROM customer_churn
5845 GROUP BY PreferredOrderCat
5846 ORDER BY AvgCashback DESC
5847 LIMIT 3;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
PreferredOrderCat	AvgCashback				
Others	304.4545				
Grocery	266.2366				
Fashion	210.4031				

➤ Find the preferred payment modes of customers whose average tenure is 10 months and have placed more than 500 orders.

```
5849 • SELECT PreferredPaymentMode
5850 FROM customer_churn
5851 GROUP BY PreferredPaymentMode
5852 HAVING AVG(Tenure)=10
5853 AND SUM(OrderCount)>500;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

PreferredPaymentMode

Result Grid

➤ Categorize customers based on their distance from the warehouse to home such as 'Very Close Distance' for distances <=5km, 'Close Distance' for <=10km, 'Moderate Distance' for <=15km, and 'Far Distance' for >15km. Then, display the churn status breakdown for each distance category.

```
5857 CASE
5858 WHEN WarehouseToHome<=5 THEN 'Very Close Distance'
5859 WHEN WarehouseToHome<=10 THEN 'Close Distance'
5860 WHEN WarehouseToHome<=15 THEN 'Moderate Distance'
5861 ELSE 'Far Distance'
5862 END AS DistanceCategory,
5863 ChurnStatus,
5864 COUNT(*) AS Customers
5865 FROM customer_churn
5866 GROUP BY DistanceCategory, ChurnStatus;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

DistanceCategory	ChurnStatus	Customers
Close Distance	Churned	265
Far Distance	Churned	498
Moderate Distance	Churned	184
Close Distance	Active	1696
Moderate Distance	Active	1106

➤ List the customer's order details who are married, live in City Tier-1, and their order counts are more than the average number of orders placed by all customers.

```
5869 • SELECT *
5870 FROM customer_churn
5871 WHERE MaritalStatus='Married'
5872 AND CityTier=1
5873 AND OrderCount >
5874 (
5875 SELECT AVG(OrderCount)
5876 FROM customer_churn
5877 );
```

Result Grid

	CustomerID	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HoursSpentOnApp	NumberOfDe
▶	50049	3	Computer	1	15	Credit Card	Male	3	3
	50119	13	Mobile Phone	1	30	Cash on Delivery	Male	3	3
	50132	15	Mobile Phone	1	16	Credit Card	Female	2	4
	50152	2	Mobile Phone	1	28	Debit Card	Female	3	2

customer_churn 27 x Apply

➤ a) Create a 'customer_returns' table in the 'ecomm' database and insert the following data:

```
5879 • CREATE TABLE customer_returns
5880 (
5881 ReturnID INT PRIMARY KEY,
5882 CustomerID INT,
5883 ReturnDate DATE,
5884 RefundAmount DECIMAL(10,2)
5885 );
```

```

5887 • INSERT INTO customer_returns
5888 VALUES
5889 (1001,50022,'2023-01-01',2130),
5890 (1002,50316,'2023-01-23',2000),
5891 (1003,51099,'2023-02-14',2290),
5892 (1004,52321,'2023-03-08',2510),
5893 (1005,52928,'2023-03-20',3000),
5894 (1006,53749,'2023-04-17',1740),
5895 (1007,54206,'2023-04-21',3250),
5896 (1008,54838,'2023-04-30',1990);

```

Result Grid				
Filter Rows:				
Edit:				
Export/Import:				
Wrap Cell Content:				
ReturnID	CustomerID	ReturnDate	RefundAmount	
1001	50022	2023-01-01	2130.00	
1002	50316	2023-01-23	2000.00	
1003	51099	2023-02-14	2290.00	
1004	52321	2023-03-08	2510.00	
1005	52928	2023-03-20	3000.00	

customer_returns 28 x

b) Display the return details along with the customer details of those who have churned and have made complaints.

```

5901 ✖ SELECT *
5902 FROM customer_returns cr
5903 INNER JOIN customer_churn cc
5904 ON cr.CustomerID = cc.CustomerID
5905 WHERE cc.ChurnStatus = 'Churned'
5906 AND cc.ComplaintReceived = 'Yes';

```

Result Grid										
Filter Rows:										
Export:										
Wrap Cell Content:										
ReturnID	CustomerID	ReturnDate	RefundAmount	CustomerID	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPayr	
1002	50316	2023-01-23	2000.00	50316	0	Computer	2	29	UPI	
1004	52321	2023-03-08	2510.00	52321	18	Mobile Phone	3	19	E wallet	
1006	53749	2023-04-17	1740.00	53749	1	Mobile Phone	3	31	Credit Card	